A Hierarchical Centered Dirichlet Process Mixture for Error Distributions in Bayesian Data Fusion under Unmeasured Confounding

Tijn Jacobs

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Abstract

We extend the Bayesian nonparametric framework for combining randomized and real-world data under unmeasured confounding by allowing the residual distribution in the accelerated failure time outcome model to differ across data sources. Building on the centered Dirichlet process (CDP) construction of Yang et al. (2010), used in a single-source AFT–BART setting by Henderson et al. (2020), we place a hierarchical Dirichlet process (HDP) prior (Teh et al., 2006) on the source-specific mixing measures, with shared atoms but source-specific weights. The mean-zero constraint required for identifiability of the regression components is enforced separately within each source by recentering. We derive a blocked Gibbs sampler that preserves conjugacy with the BART updates of the prognostic, treatment-effect, and confounding-function components.

1 Motivation

In our framework for combining randomized and real-world data under unmeasured confounding (Jacobs, 2026), the conditional expectation of the observed log-survival outcome admits the decomposition

$$\mathbb{E}[Y \mid A, X, S] \;=\; m_0(X, S) + \tau(X)\, A + (1 - S)A\, c(X), \tag{1}$$

under SUTVA, randomization in the RCT, and transportability of potential outcomes. The corresponding individual-level model takes the form

$$Y_i \;=\; m_0(X_i, S_i) + A_i\, \tau(X_i) + A_i(1 - S_i)\, c(X_i) + W_i, \qquad W_i \sim F_{S_i}, \tag{2}$$

where F_s is the error distribution within data source $s \in \{0, 1\}$ ($s = 1$ for the RCT, $s = 0$ for the real-world data). For m_0, τ, and c to be identified as the conditional-mean components defined in our framework, each F_s must have mean zero.

Henderson et al. (2020) model the residual distribution in a single-source AFT model as a location mixture of normals with a centered Dirichlet process prior on the mixing measure, leveraging the construction of Yang et al. (2010). In their setting, a single CDP is sufficient because there is only one source. In ours, the natural question is whether F_0 and F_1 should be assumed equal. We argue they should not: real-world data and randomized trials typically differ in measurement quality, follow-up protocols, and the prevalence of extreme outcomes, and these differences plausibly enter the error distribution rather than the structural mean. At the same time, the two error distributions are unlikely to be entirely unrelated, and pooling information across them through a shared base measure is desirable.

A hierarchical Dirichlet process (Teh et al., 2006) resolves this tension: F_0 and F_1 share a common pool of latent "residual types" through a top-level random measure, while admitting source-specific weights. Combined with the centering construction of Yang et al. (2010), applied within each source, this yields source-specific, mean-zero, nonparametric error distributions that borrow information through their shared atoms.

2 Notation and observed data

We observe data $\mathcal{D} = \{(Y_i^{\text{obs}}, \delta_i, A_i, X_i, S_i) : i = 1, \ldots, n\}$, where $Y_i^{\text{obs}} = \min(Y_i, \log C_i)$ is the log of the observed time, $\delta_i = \mathbb{1}\{Y_i \leq \log C_i\}$ is the event indicator, $A_i \in \{0, 1\}$ is treatment, $X_i \in \mathbb{R}^p$ is a vector of observed covariates, and $S_i \in \{0, 1\}$ is the source indicator. Write $n_s = \sum_i \mathbb{1}\{S_i = s\}$.

The model is (2). The latent complete-data log-survival time Y_i equals Y_i^{obs} if $\delta_i = 1$ and is unobserved if $\delta_i = 0$, in which case we know $Y_i > \log C_i$.

3 Error model

3.1 Per-source location mixture of Gaussians

Conditional on the source $S_i = s$, we model the residual as a location mixture of Gaussians with common scale $\sigma > 0$ and source-specific mixing measure G_s:

$$W_i \mid S_i = s, G_s, \sigma \;\sim\; f_s(w) \;=\; \int \frac{1}{\sigma}\, \phi\!\left(\frac{w - \theta}{\sigma}\right) dG_s(\theta). \tag{3}$$

Introducing a latent location θ_i for each observation gives the hierarchical representation

$$W_i \mid \theta_i, \sigma^2 \;\sim\; N(\theta_i, \sigma^2), \qquad \theta_i \mid G_{S_i} \;\sim\; G_{S_i}. \tag{4}$$

The mean-zero constraint $\mathbb{E}[W_i \mid S_i = s, G_s, \sigma] = 0$ is enforced by requiring $\int \theta \, dG_s(\theta) = 0$ for both s.

A shared scale across sources is the most parsimonious choice and parallels Henderson et al. (2020); a source-specific σ_s requires only a one-line change and is discussed in Section 8.

3.2 Hierarchical Dirichlet process prior on the mixing measures

We place a hierarchical Dirichlet process prior (Teh et al., 2006) on (G_0^*, G_1^*) via a top-level random measure G^*:

$$G^* \sim \mathrm{DP}(\gamma, H), \tag{5}$$
$$G_s^* \mid G^*, M_s \sim \mathrm{DP}(M_s, G^*), \qquad s \in \{0, 1\}, \tag{6}$$

where $H = N(0, \sigma_\theta^2)$ is the base distribution on $\mathbb{R}$, $\gamma > 0$ is the top-level concentration, and $M_0, M_1 > 0$ are source-specific concentrations. We refer to G_s^* as the *unconstrained* version of G_s; the centering described in Section 3.3 produces G_s.

By Sethuraman's stick-breaking representation (Sethuraman, 1994), the top-level measure can be written as

$$G^* \;=\; \sum_{k=1}^{\infty} \beta_k \, \delta_{\theta_k^*}, \qquad \theta_k^* \overset{\text{iid}}{\sim} H, \qquad \beta_k = u_k \prod_{l<k}(1-u_l), \qquad u_k \overset{\text{iid}}{\sim} \text{Beta}(1,\gamma). \tag{7}$$

Conditional on G^*, each G_s^* shares the same atoms but has its own weights:

$$G_s^* \;=\; \sum_{k=1}^{\infty} \pi_{sk} \, \delta_{\theta_k^*}, \tag{8}$$

with the source-specific weight vector $\boldsymbol{\pi}_s = (\pi_{sk})_{k\geq 1}$ distributed according to the conditional law $\text{DP}(M_s, G^*)$ projected onto the atoms.

3.3 Mean-zero centering

For each source s, define the mean of the unconstrained measure:

$$\mu_s \;=\; \int \theta \, dG_s^*(\theta) \;=\; \sum_{k=1}^{\infty} \pi_{sk} \, \theta_k^*. \tag{9}$$

Following Yang et al. (2010), the *centered* measure is

$$G_s \;=\; \sum_{k=1}^{\infty} \pi_{sk} \, \delta_{\theta_k^* - \mu_s}. \tag{10}$$

By construction $\int \theta \, dG_s(\theta) = 0$ almost surely, so (3) satisfies $\mathbb{E}[W_i \mid S_i = s, G_s, \sigma] = 0$ as required.

Although the unconstrained atoms θ_k^* are shared between sources, the centered atoms $\theta_{sk} := \theta_k^* - \mu_s$ are source-specific: they share their unconstrained location θ_k^* but are shifted by source-specific amounts μ_0 and μ_1. This is the precise sense in which the two error distributions borrow information while retaining flexibility: the *shapes* of F_0 and F_1 are linked through the shared θ_k^*, while the *weights* on those shapes (and the resulting mean shifts) are source-specific.

3.4 Truncation for posterior computation

For computation we truncate the top-level stick-breaking at a finite level K (Ishwaran and James, 2001): $u_K = 1$, so $\beta_K = 1 - \sum_{k<K} \beta_k$ and $\beta_k = 0$ for $k > K$. Conditional on the truncated $\boldsymbol{\beta} \in \Delta^{K-1}$ we use the finite-dimensional approximation

$$\boldsymbol{\pi}_s \mid \boldsymbol{\beta}, M_s \;\sim\; \text{Dirichlet}(M_s\beta_1, \ldots, M_s\beta_K), \tag{11}$$

which is the standard truncated representation of the HDP (Teh et al., 2006; Wang et al., 2009); the approximation becomes exact as $K \to \infty$. We monitor the largest occupied component index across iterations and increase K if it approaches the truncation bound (default $K = 50$, following Henderson et al. (2020)).

4 Full hierarchical model

Introducing latent cluster assignments $Z_i \in \{1, \dots, K\}$ such that $\theta_i = \theta^*_{Z_i} - \mu_{S_i}$, the full hierarchical model is:

$$
\begin{aligned}
Y_i \mid Z_i, \boldsymbol{\theta}^*, \sigma^2, m_0, \tau, c &\sim N\big(m_0(X_i, S_i) + A_i \tau(X_i) + A_i(1 - S_i)\, c(X_i) + \theta^*_{Z_i} - \mu_{S_i},\ \sigma^2\big), && (12)\\
Z_i \mid \boldsymbol{\pi}_{S_i} &\sim \text{Categorical}(\boldsymbol{\pi}_{S_i}), && (13)\\
\boldsymbol{\pi}_s \mid \boldsymbol{\beta}, M_s &\sim \text{Dirichlet}(M_s\beta_1, \dots, M_s\beta_K), \qquad s \in \{0, 1\}, && (14)\\
\boldsymbol{\beta} \mid \gamma &\sim \text{Stick}_K(\gamma), && (15)\\
\theta^*_k &\overset{\text{iid}}{\sim} N(0, \sigma^2_\theta), \qquad k = 1, \dots, K, && (16)\\
m_0, \tau, c &\sim \text{BART}, && (17)\\
\sigma^2 &\sim \kappa\nu/\chi^2_\nu, && (18)\\
\gamma &\sim \text{Gamma}(a_\gamma, b_\gamma), && (19)\\
M_s &\sim \text{Gamma}(a_M, b_M), \qquad s \in \{0, 1\}, && (20)
\end{aligned}
$$

with $\mu_s = \sum_k \pi_{sk}\theta^*_k$ a deterministic function of $\boldsymbol{\pi}_s$ and $\boldsymbol{\theta}^*$. Here $\text{Stick}_K(\gamma)$ denotes the truncated stick-breaking distribution with $\text{Beta}(1, \gamma)$ breaks.

For the BART components we follow Chipman et al. (2010) with the AFT-specific adaptations of Henderson et al. (2020): $J = 200$ trees, $\alpha = 0.95$, $\beta = 2$, response transformation $y^{\text{tr}}_i = y_i \exp(-\hat{\mu}_{\text{AFT}})$ based on a preliminary parametric log-normal AFT fit, and calibration of κ, σ^2_θ via the Yamato-type approximation described in Henderson et al. (2020, Section 2.4). For the concentration hyperpriors we set $a_\gamma = a_M = 2$, $b_\gamma = b_M = 0.1$, mirroring Henderson's defaults; the resulting prior on each concentration has mean 20 and prior mode 10.

5 Posterior computation via blocked Gibbs sampling

Let $\boldsymbol{\Theta}$ denote the full state of the sampler:

- BART parameters (trees and node values) for m_0, τ, c;
- Cluster assignments $\boldsymbol{Z} = (Z_1, \dots, Z_n)$;
- Unconstrained atoms $\boldsymbol{\theta}^* = (\theta^*_1, \dots, \theta^*_K)$;
- Top-level stick parameters $\boldsymbol{u}$ (equivalently $\boldsymbol{\beta}$);
- Source-specific weights $\boldsymbol{\pi}_0, \boldsymbol{\pi}_1$;
- Concentration parameters γ, M_0, M_1;
- Error scale σ^2;
- Imputed log-survival times $\{Y_i : \delta_i = 0\}$.

The source means $\boldsymbol{\mu} = (\mu_0, \mu_1)$ are recomputed deterministically as $\mu_s = \sum_k \pi_{sk}\theta^*_k$ whenever $\boldsymbol{\pi}$ or $\boldsymbol{\theta}^*$ change.

For notational economy, define the *partial residual* for individual i:

$$R_i \ := \ Y_i - m_0(X_i, S_i) - A_i\, \tau(X_i) - A_i(1 - S_i)\, c(X_i). \tag{21}$$

Conditional on the BART components and $Z_i = k$, we have $R_i \sim N(\theta^*_k - \mu_{S_i}, \sigma^2)$.

One sweep of the Gibbs sampler updates each block in turn. We describe the steps in the order in which they should be executed.

5.1 Step 1: Data augmentation for censored observations

For each i with $\delta_i = 0$, draw the imputed log-survival time

$$Y_i \mid \cdot \;\sim\; \text{TruncatedNormal}(\eta_i,\, \sigma^2;\, [\log C_i, \infty)), \tag{22}$$

where $\eta_i = m_0(X_i, S_i) + A_i\tau(X_i) + A_i(1 - S_i)\, c(X_i) + \theta^*_{Z_i} - \mu_{S_i}$ is the current conditional mean of Y_i.

5.2 Step 2: BART backfitting

The three BART components are updated sequentially via Bayesian backfitting (Chipman et al., 2010). For each component, the working response is the partial residual obtained by subtracting the contributions of all other components from Y_i, along with the current cluster shift $\theta^*_{Z_i} - \mu_{S_i}$.

Update m_0. Working response:

$$R_i^{(m_0)} \;=\; Y_i - A_i\, \tau(X_i) - A_i(1 - S_i)\, c(X_i) - \big(\theta^*_{Z_i} - \mu_{S_i}\big). \tag{23}$$

Treat $R_i^{(m_0)} \sim N(m_0(X_i, S_i), \sigma^2)$ and apply the standard MH tree moves and conjugate node-value updates on the predictor vector (X_i, S_i).

Update τ. Active only for treated observations ($A_i = 1$). Working response:

$$R_i^{(\tau)} \;=\; Y_i - m_0(X_i, S_i) - (1 - S_i)\, c(X_i) - \big(\theta^*_{Z_i} - \mu_{S_i}\big), \qquad i \text{ with } A_i = 1. \tag{24}$$

Treat $R_i^{(\tau)} \sim N(\tau(X_i), \sigma^2)$ and update the BART for τ on this subset. Control observations carry no information about τ.

Update c. Active only for treated observations in the RWD ($A_i = 1, S_i = 0$). Working response:

$$R_i^{(c)} \;=\; Y_i - m_0(X_i, 0) - \tau(X_i) - \big(\theta^*_{Z_i} - \mu_0\big), \qquad i \text{ with } A_i = 1, S_i = 0. \tag{25}$$

Treat $R_i^{(c)} \sim N(c(X_i), \sigma^2)$ and update the BART for c on this subset.

After updating all three BART components, recompute R_i as in (21) for use in the remaining steps.

5.3 Step 3: Cluster assignments

For each i, sample $Z_i \in \{1, \ldots, K\}$ from

$$P(Z_i = k \mid \cdot) \;\propto\; \pi_{S_i,k}\, \phi\left(\frac{R_i - (\theta^*_k - \mu_{S_i})}{\sigma}\right), \qquad k = 1, \ldots, K. \tag{26}$$

5.4 Step 4: Unconstrained atoms

Following Henderson et al. (2020), atoms are updated in the unconstrained parameterization and the centering is reapplied deterministically afterwards. Let $n_k = \sum_i \mathbb{1}\{Z_i = k\}$ be the total number of observations in cluster k across both sources, and define the cluster sum $\bar{R}_k = \sum_{i:Z_i=k} R_i$.

The conjugate Gaussian update is

$$\theta_k^* \mid \cdot \;\sim\; N\left(\frac{\sigma^{-2}\,\bar{R}_k}{\sigma_\theta^{-2} + n_k\,\sigma^{-2}},\; \frac{1}{\sigma_\theta^{-2} + n_k\,\sigma^{-2}}\right), \qquad k = 1, \dots, K. \tag{27}$$

After updating all θ_k^*, recompute the source means and centered atoms:

$$\mu_s \;=\; \sum_{k=1}^{K} \pi_{sk}\,\theta_k^*, \qquad \theta_{sk} \;=\; \theta_k^* - \mu_s, \qquad s \in \{0, 1\}. \tag{28}$$

Remark. The update (27) is the standard Gibbs step for the unconstrained mixing measure; the centering acts as a deterministic transformation between an unidentified parameterization $(m_0^{\mathrm{unc}}, \boldsymbol{\theta}^*)$ and the identified $(m_0, \boldsymbol{\theta})$. As discussed by Yang et al. (2010) and Henderson et al. (2020), this parameter-expansion strategy preserves the correct marginal posterior for the centered quantities.

5.5 Step 5: Top-level stick weights

The HDP update for the top-level stick-breaking weights uses the auxiliary table-count augmentation of Teh et al. (2006), originally derived from Antoniak (1974).

Step 5a: Sample table counts. Let $n_{sk} = \sum_i \mathbb{1}\{S_i = s, Z_i = k\}$. For each (s, k) with $n_{sk} > 0$, sample the auxiliary table count $t_{sk} \in \{1, \dots, n_{sk}\}$. The marginal distribution is

$$P(t_{sk} = t \mid \cdot) \;\propto\; |s(n_{sk}, t)|\,(M_s\beta_k)^t, \tag{29}$$

where $|s(n, t)|$ are unsigned Stirling numbers of the first kind. In practice t_{sk} is generated via the equivalent representation

$$t_{sk} \;=\; \sum_{j=1}^{n_{sk}} B_{sk,j}, \qquad B_{sk,j} \;\sim\; \text{Bernoulli}\left(\frac{M_s\beta_k}{M_s\beta_k + j - 1}\right), \tag{30}$$

which avoids the Stirling-number evaluation. Set $t_{sk} = 0$ if $n_{sk} = 0$.

Step 5b: Sample stick weights. Let $t_{\cdot k} = t_{0k} + t_{1k}$. Sample the truncated sticks

$$u_k \mid \cdot \;\sim\; \text{Beta}\left(1 + t_{\cdot k},\; \gamma + \sum_{l>k} t_{\cdot l}\right), \qquad k = 1, \dots, K - 1, \tag{31}$$

and set $u_K = 1$. Recompute $\beta_k = u_k \prod_{l<k}(1 - u_l)$ for $k = 1, \dots, K$.

5.6 Step 6: Source-specific weights

Given $\boldsymbol{\beta}$ and the cluster assignments, $\boldsymbol{\pi}_s$ has a conjugate Dirichlet update:

$$\boldsymbol{\pi}_s \mid \cdot \;\sim\; \text{Dirichlet}(M_s\beta_1 + n_{s1},\, \dots,\, M_s\beta_K + n_{sK}), \qquad s \in \{0,1\}. \tag{32}$$

After this step, recompute μ_s and $\theta_{sk} = \theta_k^* - \mu_s$.

5.7 Step 7: Concentration parameters

Both γ and M_s are updated using the auxiliary-variable approach of Escobar and West (1995).

Update γ. Let K^* denote the number of occupied top-level components and let $t_{\cdot\cdot} = \sum_{s,k} t_{sk}$ denote the total number of tables across both restaurants.

1. Sample $\eta_\gamma \mid \gamma, t_{\cdot\cdot} \sim \text{Beta}(\gamma + 1, t_{\cdot\cdot})$.
2. Sample $\gamma \mid \eta_\gamma, K^*$ from the two-component mixture

$$\begin{aligned}\gamma \mid \cdot \sim \omega_\gamma\, &\text{Gamma}(a_\gamma + K^*,\; b_\gamma - \log\eta_\gamma) \\ &+ (1-\omega_\gamma)\,\text{Gamma}(a_\gamma + K^* - 1,\; b_\gamma - \log\eta_\gamma),\end{aligned}$$

   with

$$\frac{\omega_\gamma}{1-\omega_\gamma} \;=\; \frac{a_\gamma + K^* - 1}{t_{\cdot\cdot}(b_\gamma - \log\eta_\gamma)}.$$

Update M_s. For each $s \in \{0,1\}$, with $t_{s\cdot} = \sum_k t_{sk}$ source-specific table counts:

1. Sample $\eta_{M,s} \mid M_s, n_s \sim \text{Beta}(M_s + 1, n_s)$.
2. Sample $M_s \mid \eta_{M,s}, t_{s\cdot}$ from the analogous two-component Gamma mixture with $t_{s\cdot}$ in place of K^* and n_s implicit in $\eta_{M,s}$.

See Teh et al. (2006, Appendix A) for the full derivation in the HDP setting.

5.8 Step 8: Error scale

Conditional on assignments, atoms, and BART components,

$$\sigma^2 \mid \cdot \;\sim\; \text{InverseGamma}\!\left(\frac{\nu+n}{2},\, \frac{\kappa\nu + \hat{s}^2}{2}\right), \qquad \hat{s}^2 = \sum_{i=1}^{n} \big(R_i - (\theta^*_{Z_i} - \mu_{S_i})\big)^2. \tag{33}$$

5.9 Algorithmic summary

One sweep of the Gibbs sampler consists of:

(1) Impute Y_i for censored i (Step 1).
(2) Update the BART components m_0, then τ, then c, recomputing R_i at the end (Step 2).
(3) Update cluster assignments $\boldsymbol{Z}$ (Step 3).

(4) Update the unconstrained atoms $\boldsymbol{\theta}^*$ and recompute μ_s, θ_{sk} (Step 4).
(5) Update top-level table counts $\{t_{sk}\}$ and sticks $\boldsymbol{u}$, then $\boldsymbol{\beta}$ (Step 5).
(6) Update source weights $\boldsymbol{\pi}_0, \boldsymbol{\pi}_1$ and recompute μ_s, θ_{sk} (Step 6).
(7) Update concentration parameters γ, M_0, M_1 (Step 7).
(8) Update σ^2 (Step 8).

6 Practical considerations

Truncation. The default $K = 50$ follows Henderson et al. (2020). Monitor $\max_i Z_i$ across iterations; if it equals K in more than a small fraction of iterations, K is too small.

Hyperparameter calibration. The induced prior on $\mathrm{Var}(W \mid S = s)$ under the HDP-CDP has the same structural form as in the single-source case of Henderson et al. (2020), with source-specific contributions. The Yamato-type approximation (Yamato, 1984) they use to obtain a tractable approximation for $\sum_k \pi_{sk}(\theta_k^* - \mu_s)^2/\sigma_\theta^2$ extends without modification to each source separately. Calibration to a preliminary $\hat{\sigma}_W^2$ obtained from a parametric log-normal AFT fit is recommended, with default tail probability $q = 0.5$.

Initialization. Initialize the BART components via the standard Chipman et al. (2010) initialization; assign cluster labels by k-means on residuals from an initial parametric fit; set $\boldsymbol{\beta} = (1/K, \ldots, 1/K)$, $\boldsymbol{\pi}_s = \boldsymbol{\beta}$, $\gamma = M_0 = M_1 = 1$.

Diagnostics. The source-specific means μ_0, μ_1 and the variances $\mathrm{Var}(W \mid S = s)$ are derived quantities of the posterior and serve as natural diagnostics of how much source-specific structure the HDP is exploiting. Posterior summaries of $|\mu_0 - \mu_1|$ and of the source-specific residual variances communicate the extent to which F_0 and F_1 differ in their first two moments; the full residual densities $\hat{f}_s(w) = K^{-1} \sum_k \pi_{sk}\, \phi((w - \theta_{sk})/\sigma)/\sigma$ can be plotted for direct visual comparison.

7 Implementation notes (FusionForests R package)

The model above is partially implemented in the `FusionForests` R package as of this writing. This section documents what is implemented, the hyperparameter choices baked into the code, and the things to be careful about when describing those choices in the eventual paper. Anything not listed here either matches the description above verbatim or is flagged at the end as future work.

7.1 Implemented modes

A single user-facing argument `error_dist` selects one of five residual priors. One is the standard normal baseline; the other four span the design space from a pooled DP through to the full HDP-CDP of Section 3.3:

`gaussian` $W_i \sim N(0, \sigma^2)$; no mixture; the existing inverse-χ^2 update for σ is used.

`shared_dp` One centred DP pooled across both sources. Global recentre $\theta_k \mapsto \theta_k - \sum_k \pi_k \theta_k$. Direct port of Henderson et al. (2020).

`source_dp` Independent centred DPs per source (G_0 and G_1 with their own atoms, weights, and concentration), shared scale σ. Per-source recentre.

`source_dp_scale` Same as `source_dp` but with per-source scales σ_0, σ_1 given an inverse-χ^2 conjugate update each sweep. Per-observation precision is propagated to the BART back-fitting via per-observation precision weights (see §7.3); the BCF treatment-effect weight b_i^2 is composed multiplicatively.

`source_hdp` Full hierarchical centred Dirichlet process (Stage B). Atoms θ_k^* are shared across sources via a top-level DP(γ, H); per-source weights π_{sk} and means μ_s restore identifiability of the structural components. Top-level sticks β are updated via Antoniak–Teh table-count augmentation; the per-source concentrations M_s and the top-level γ via Escobar–West two-component Gamma mixtures. Implementation details in §7.5.

The Stage A modes (`source_dp`, `source_dp_scale`) correspond to the "independent DPs" stepping stone discussed in the implementation plan; `source_hdp` is the full HDP-CDP of Sections 3.3–6. A combined HDP-with-per-source-scale mode is not provided; if you need it, layer the `source_dp_scale`-style σ_s update on top of the HDP path (a few lines, but currently out of scope).

7.2 Hyperparameters and how they are set

The relevant hyperparameters and their defaults are summarised in Table 1. All values listed live on the *standardised* response scale: the package centres y and divides by an initial $\hat{\sigma}$ derived either from sd(y) (continuous outcomes) or from a parametric AFT fit (survival outcomes), and the DP atoms and per-source σ_s are reported back on the original response scale by undoing this transformation at the end.

Symbol	Default	R-level control
K (truncation level)	50	`error_truncation_K`
σ_θ (atom prior SD)	0.5	`error_atom_scale`
α initial value	1	`error_mass_init`
(ψ_1, ψ_2) on α	$(2.0, 0.1)$	not exposed
ν, λ on σ^2 (and σ_s^2)	inherited from the existing `nu`, `q`	via existing `nu`, `q`

Table 1: Hyperparameters in the residual-mixture prior. Defaults are chosen to match Henderson et al. (2020) where applicable. Values that should be reported in the paper are flagged in the text below.

Truncation K. Default 50. Henderson et al. (2020) uses 50 in their two-arm variant and 200 in their single-arm variant; the difference does not affect their posterior summaries, suggesting 50 is comfortable for moderate n. In our simulations under $n_{\mathrm{RCT}} = 100$, $n_{\mathrm{RWD}} = 200$ the number of occupied components rarely exceeds 20, with the RCT-only DP under `source_dp` occupying roughly 18/50 on average. *For the paper:* report posterior-mean occupancy by group and verify it stays well below K.

Atom prior scale σ_θ. The atoms have prior $\theta_k \sim N(0, \sigma_\theta^2)$. The default $\sigma_\theta = 0.5$ on the standardised scale lets atoms reach roughly ± 1 standard residual unit with prior probability ≈ 0.95. Henderson et al. (2020) calibrate this from a preliminary parametric fit via their `FindKappa` helper, which after standardisation gives $\sigma_\theta^2 \approx \sigma_\theta^2 / Q$ with Q a chi-square-derived quantile – numerically similar to our flat default. *For the paper:* either match Henderson's calibration (port `FindKappa`) or document that the standardised-scale default is robust and report posterior atom locations to justify it. If atoms cluster against the prior support boundary, the prior is too tight.

Concentration prior (ψ_1, ψ_2). The conjugate update sets $\alpha \mid \cdot \sim \text{Gamma}(\psi_1 + K - 1,\ \psi_2 - \sum_h \log(1 - V_h))$, so the prior is $\alpha \sim \text{Gamma}(\psi_1, \psi_2)$ with prior mean $\psi_1/\psi_2 = 20$. These constants are inherited verbatim from AFTrees and are *not* exposed at the R level. *For the paper:* either expose them and report a sensitivity analysis (smaller ψ_1 shrinks α toward 0, fewer effective components) or justify the AFTrees default. In our simulations the posterior α varied from ≈ 2 (sharp clusters) to ≈ 24 (diffuse), so the default does not dominate the data.

Mass initialisation. The R argument `error_mass_init` sets $\alpha^{(0)} = 1$ by default; AFTrees uses 10. The conjugate update converges within a few sweeps regardless, and we have not observed sensitivity to this choice. Drop from the paper.

ν, λ for σ_s^2. In `source_dp_scale` the per-source scale update is $\sigma_s^2 \mid \cdot \sim \text{IG}((\nu + n_s)/2,\ (\nu\lambda + SS_s)/2)$ where $SS_s = \sum_{i:S_i=s}(\varepsilon_i - \theta_{Z_i})^2$. The hyperparameters (ν, λ) are inherited from the outer Gibbs sampler ($\nu = 3$ by default, λ derived from $\hat{\sigma}^2 \cdot \chi^2_{\nu,1-q}/\nu$ with $q = 0.9$), *shared* across sources. *For the paper:* consider allowing source-specific (ν_s, λ_s), calibrated independently from the RCT and RWD residuals of a preliminary parametric fit. Sharing is principled when RCT and RWD are believed to have similar prognostic noise, but the implementation cost of stratifying is one line.

Numerical caps. Both the global σ and the per-source σ_s updates cap the draw at 10 (on the standardised scale) to guard against pathological early-burn-in draws. This mirrors the existing `UpdateSigma`. *For the paper:* this is a numerical detail and need not be mentioned, but the cap can mask divergence – the diagnostic is to inspect posterior σ traces.

7.3 Forest backfitting under per-source σ_s

In `source_dp_scale`, the joint likelihood has per-observation variance $\sigma_{S_i}^2$. The BART forests are passed a reference scale $\sigma_{\text{ref}} = \sqrt{(n_0\sigma_0^2 + n_1\sigma_1^2)/n}$ together with per-observation precision weights

$$w_i \ = \ (\sigma_{\text{ref}}/\sigma_{S_i})^2 \cdot w_i^{(0)},$$

where $w_i^{(0)}$ is the existing weight (either b_i^2 for the τ and c forests under BCF, or 1 for the others). The effective per-observation variance the forest sees is then $\sigma_{\text{ref}}^2/w_i = \sigma_{S_i}^2/w_i^{(0)}$ – the desired per-source bandwidth, with the BCF reweighting preserved. The weights are recomputed every sweep after the DP block and exposed to the forests via the existing `SetWeights` API (which stores a pointer, so in-place updates suffice). In `source_dp_scale`, the *global* σ in the outer Gibbs is replaced by σ_{ref} and the existing `UpdateSigma` step is skipped; the per-source σ_s updates inside `MixtureDP` drive the scale entirely.

7.4 Centring is post-hoc renormalisation

The mean-zero constraint $\sum_k \pi_{sk}\theta_{sk} = 0$ is enforced as a *deterministic* recentre after each atom block, exactly as in Yang et al. (2010) and AFTrees: atoms are sampled from the unconstrained posterior, then shifted by $-\mu_s = -\sum_k \pi_{sk}\theta^*_k$. This is a parameter-expansion in the sense of ?: the implied prior on the centred atoms is the marginal of the unconstrained prior conditional on the centring constraint, which is preserved by the deterministic transformation. For `source_dp`, `source_dp_scale`, and `source_hdp`, the recentre is *per source*; for `shared_dp`, it is global, which couples the RCT and RWD likelihoods through the shared μ.

7.5 Full HDP-CDP implementation (`source_hdp`)

The HDP-CDP mode adds the following state on top of the Stage A per-source containers:

- `locations_shared_`: vector of length K, the unconstrained shared atoms θ^*_k.
- `beta_`: vector of length K, the top-level stick weights β_k.
- `gamma_top_`: scalar, the top-level concentration γ.
- `mu_g_`: per-source means $\mu_s = \sum_k \pi_{sk}\theta^*_k$, recomputed deterministically.
- `table_counts_`: matrix t_{sk} for $(s,k) \in \{0,1\} \times \{1,\ldots,K\}$.

The existing per-source `locations_[g]` container holds the *centred* atoms $\theta^*_k - \mu_g$, refreshed each sweep so that downstream consumers (`GetIndividualShifts` for the BART working response, posterior storage of `dp_locations`) need no special-casing.

One Gibbs sweep in `source_hdp` runs these steps in order:

(1) `updateLabelsGroup` for each g: identical to Stage A but reads the per-source centred atoms from `locations_[g]`.
(2) `tabCountsGroup` for each g: populate n_{sk}.
(3) `updateBeta` (Antoniak–Teh):
  - For each (s,k) with $n_{sk} > 0$, sample $t_{sk} = \sum_{j=1}^{n_{sk}} B_j$ with $B_j \sim \mathrm{Bern}\big(M_s\beta_k/(M_s\beta_k + j - 1)\big)$. We floor at $t_{sk} = 1$ to enforce "at least one table per occupied component", matching Teh et al. (2006).
  - Stick-breaking: $u_h \sim \mathrm{Beta}(1 + t_{\cdot h}, \gamma + \sum_{l>h} t_{\cdot l})$, $\beta_h = u_h \prod_{l<h}(1 - u_l)$; closing slot β_K as the residual.
(4) `updateMixHDP` for each g: $\boldsymbol{\pi}_s \sim \mathrm{Dirichlet}(M_s\beta_k + n_{sk})$. We use the log-Dirichlet helper in `Random.h` for numerical safety on small $M_s\beta_k + n_{sk}$.
(5) `updateMassHDP` for each g (Escobar–West): $\eta_{M,s} \sim \mathrm{Beta}(M_s + 1, n_s)$, then M_s from a two-component Gamma mixture with mixture weight $\omega_{M,s}$ driven by the table totals $t_{s\cdot}$.
(6) `updateAtomsShared`: conjugate Gaussian per atom, pooling observations across sources. For cluster k with $n_k = n_{0k} + n_{1k}$ and "uncentred" sum $\sum_{i:Z_i=k}(R_i + \mu_{S_i})$ (the model says $R_i = \theta^*_{Z_i} - \mu_{S_i} + \sigma\varepsilon_i$, so the augmented quantity is Gaussian around θ^*_k), the posterior is the standard Gaussian-Gaussian conjugate with prior $\theta^*_k \sim N(0, \sigma^2_\theta)$.
(7) `recomputeCenteredLocations`: $\mu_g \leftarrow \sum_k \pi_{gk}\theta^*_k$; `locations_[g][k]` $\leftarrow \theta^*_k - \mu_g$.
(8) `updateGamma` (Escobar–West): $\eta_\gamma \sim \mathrm{Beta}(\gamma+1, t_{\cdot\cdot})$, then γ from a two-component Gamma mixture driven by $K^* = \#\text{components with } t_{\cdot k} > 0$.

Hyperparameters. The top-level γ inherits the same $\mathrm{Gamma}(\psi_1 = 2.0,\ \psi_2 = 0.1)$ prior as the per-source M_s (Table 1), matching the defaults in Teh et al. (2006) and Section 6. The atom prior $\theta_k^* \sim N(0, \sigma_\theta^2)$ uses the same `error_atom_scale` as the other modes. No new R-level controls are exposed in HDP mode.

Numerical guards. The Beta draws for stick-breaking and Escobar–West are clamped strictly inside $(0, 1)$ via `nextafter`; the Gamma rate $\psi_2 - \log \eta$ is floored at 10^{-10}; the mixture weight $\omega = \text{odds}/(1 + \text{odds})$ is clamped to $[0, 1]$. The atom posterior is unbounded by construction since the conjugate update is straightforward Gaussian.

Forest backfitting in HDP mode. The per-observation working response uses the centred atom $\theta_{Z_i}^* - \mu_{S_i}$ exactly as in `source_dp`; the BART backfit, σ update, and censored augmentation are untouched. The shared atoms enter the model *only* through the cluster-shift subtracted in the working response.

7.6 Empirical pathology of `shared_dp`

In our 25-replicate simulation under bimodal RWD noise (results in `examples/NP_error_simulation_report.m` `shared_dp` produced unstable CATE RMSE with coefficient of variation ≈ 0.85 across replicates: most replicates were fine, a few were catastrophic. The mechanism is the global centring: the shared mixture must accommodate the RCT mode near 0 and the RWD modes near ± 1 simultaneously, so the centring shift μ has to balance them; with $n_{\text{RCT}} = 100$ the sampler occasionally lands in a label configuration that mis-attributes RCT observations to the wide RWD modes, biasing both the cluster shifts and the structural mean estimates. The Stage A modes (`source_dp`, `source_dp_scale`) avoid the issue by decoupling the centring entirely; `source_hdp` avoids it by recentring *per source* while still sharing the atom *positions* through the top-level G^*. *For the paper:* this is the central empirical motivation for the modelling trajectory pooled $\Rightarrow$ per-source $\Rightarrow$ HDP-CDP: each step is needed to fix a specific failure mode of the previous one.

7.7 Posterior outputs

When `error_dist` is anything other than `gaussian`, the fit object gains:

- `dp_mix_prop`: list of $N_{\text{post}} \times K$ posterior matrices (one per group).
- `dp_locations`: same shape, the centred atoms on the response scale (multiplied by $\hat{\sigma}$). For `source_hdp` these equal $\theta_k^* - \mu_g$ at each draw.
- `dp_mass`: list of N_{post}-vectors of per-source concentration draws (M_s).
- `dp_sigma_g` (`source_dp_scale` only): list of N_{post}-vectors of per-source scale draws on the response scale.
- `dp_locations_shared` (`source_hdp` only): $N_{\text{post}} \times K$ matrix of the shared atoms θ_k^* on the response scale.
- `dp_beta` (`source_hdp` only): $N_{\text{post}} \times K$ matrix of top-level stick weights (scale-free).
- `dp_gamma` (`source_hdp` only): N_{post}-vector of the top-level concentration.
- `dp_mu_g` (`source_hdp` only): list of N_{post}-vectors of per-source means μ_s on the response scale.

The diagnostics described in §6 (occupancy, $|\mu_0 - \mu_1|$, residual densities) are computable from these outputs. *For the paper:* include at least the average occupancy per group, the posterior of σ (or σ_0, σ_1), the per-source α_s / M_s, and – for HDP runs – the per-iteration table counts t_s.

(derivable from β and M_s via the auxiliary representation) as the principled effective-sample-size diagnostic of cross-source borrowing.

7.8 What is *not* yet implemented

1. A combined HDP + per-source-scale mode (i.e. `source_hdp` with per-source σ_s). Layering the `source_dp_scale` update path on top of the HDP routine is a few lines, but currently the HDP mode uses a single σ matching `source_dp`.
2. Source-specific (ν_s, λ_s) for the σ_s prior in `source_dp_scale`.
3. Concentration-prior hyperparameters (ψ_1, ψ_2) exposed at the R level (currently hard-coded Gamma$(2.0, 0.1)$ for both the per-source M_s and the top-level γ).
4. DP-aware censored-data augmentation in `source_dp_scale`: the truncated-normal step currently uses σ_{ref} rather than per-observation σ_{S_i}. For continuous outcomes this is irrelevant; for AFT survival data with heavy censoring per source it would be more faithful to use per-source scales here too.
5. Adaptive K: the truncation is currently static. An automatic tightening based on the largest occupied top-level component (i.e. the largest k with $t_{\cdot k} > 0$ in HDP mode) would be a small addition.
6. Per-iteration storage of t_{sk}: currently the table counts are recomputed each sweep but not stored. Exposing them would let the user inspect the borrowing dynamics directly, rather than indirectly via β and M_s.

7.9 Practical considerations specific to running the package

Small sources. The RCT in our simulations had $n_{\text{RCT}} = 100$ and the corresponding DP under `source_dp` occupied ~ 18 atoms – around 5–6 observations per component on average. This is the regime in which the per-source DP can over-fit, even though the CATE estimate is unaffected in our runs. If your RCT is smaller still, prefer `source_dp_scale` (which compresses redundant flexibility into the scale parameter) or set `error_atom_scale` smaller to shrink the atoms.

When to use which mode. A pragmatic decision tree, given current evidence:

- Residual diagnostics look roughly normal $\Rightarrow$ `gaussian`.
- Residuals look symmetric but heavy-tailed or noticeably non-normal, similar across sources $\Rightarrow$ `shared_dp`, but verify replicates are stable (the bimodal-RWD pathology of §7.4 can bite even on milder shape misspecification).
- Per-source residuals look qualitatively different (e.g. one looks normal, one is multimodal or strongly skewed) and the sources have similar size $\Rightarrow$ `source_dp`; add `source_dp_scale` if the residual *scales* also differ substantially.
- Per-source residuals differ in shape but the smaller source is small enough that independent atoms would over-fit (see the "Small sources" note below) $\Rightarrow$ `source_hdp`. The shared atoms regularise the small source by borrowing structure from the larger one, while per-source weights and centring keep the structural-mean estimands identified.

Reproducibility. The C++ sampler uses R's RNG state via `Rcpp::RNGScope`; set `set.seed()` immediately before calling `FusionForest`. Parallel runs over replicates should use deterministic seeds derived from the replicate index, as in `examples/NP_error_simulation.R`.

8 Extensions

Several extensions follow naturally from this construction.

Source-specific scales. Replacing σ by σ_s requires only stratifying Steps 3 and 8 by source. This is appropriate when RWD has notably different noise magnitude than the RCT.

Nested rather than hierarchical Dirichlet process. A nested Dirichlet process (Rodríguez et al., 2008) on (G_0^*, G_1^*) would cluster the residual *distributions themselves*, identifying subpopulations within the RWD whose residual structure matches the RCT and subpopulations whose does not. This is thematically aligned with the confounding-function decomposition: the bias function $c(X)$ absorbs the conditional-mean shift, and the nested DP would absorb the residual-shape shift.

Covariate-dependent error distribution. A dependent Dirichlet process (MacEachern, 1999) with covariate-dependent weights would extend the construction to fully heteroscedastic errors. Combining with the heteroskedastic AFT-BART approach of Sparapani et al. (2023), in which a second BART models $\log \sigma(X)$, yields an even more flexible alternative.

Connection to Bayesian borrowing diagnostics. Source-specific concentration parameters M_s, jointly with the source-specific table counts $t_{s\cdot}$ produced as a by-product of the sampler, function as effective-sample-size-style summaries of how much the RWD-specific residual distribution borrows from the shared base measure G^*. This connects the construction to the broader Bayesian-borrowing literature on dynamic information-sharing diagnostics (Hupf et al., 2021).

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